

Nicholas Richardson

Curriculum Vitae

Institute of Applied Mathematics
311-6356 Agricultural Road
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njericha.github.io

Research Interests

Audio & Signal Processing, Unsupervised Learning, and Tensor Factorizations

Education

- Sep. 2022 – **Doctor of Philosophy**, *University of British Columbia*, Vancouver, BC
(Apr. 2026) Institute of Applied Mathematics, supervised by Michael P. Friedlander
- Sep. 2020 – **Master of Mathematics**, *University of Waterloo*, Waterloo, ON
May 2022 Applied Mathematics Department, supervised by Giang Tran
- Sep. 2016 – **Bachelor of Mathematics**, *University of Waterloo*, Waterloo, ON
Apr. 2020 Mathematical Physics Major, Pure Mathematics Minor

Publications

- 2025 Naomi Graham, **Nicholas Richardson**, Michael P. Friedlander, and Joel Saylor. Tracing Sedimentary Origins in Multivariate Geochronology via Constrained Tensor Factorization. *Mathematical Geosciences*.
- 2023 Michael L Waite and **Nicholas Richardson**. Viscous and diffusive generation of potential vorticity in stratified turbulence. In *AGU Fall Meeting Abstracts*, volume 2023, pages NG41A–06.
- 2023 Michael L. Waite and **Nicholas Richardson**. Potential Vorticity Generation in Breaking Gravity Waves. *Atmosphere*, 14(5):881.
- 2023 **Nicholas Richardson**, Hayden Schaeffer, and Giang Tran. SRMD: Sparse Random Mode Decomposition. *Communications on Applied Mathematics and Computation*.
- 2023 Lam Si Tung Ho, **Nicholas Richardson**, and Giang Tran. Adaptive group Lasso neural network models for functions of few variables and time-dependent data. *Sampling Theory, Signal Processing, and Data Analysis*, 21(2):28.
- 2022 **Nicholas Richardson**. A Sparse Random Feature Model for Signal Decomposition. *UWSpace*. (Master's Thesis).

In Progress

Joel Saylor, **Nicholas Richardson**, Naomi Graham, Robert Lee, and Michael P. Friedlander. Tracking cu-fertile sediment sources via multivariate petrochronological mixture modelling of detrital zircons. (in review).

Nicholas Richardson, Noah Marusenko, and Michael P. Friedlander. Multiple Scale Methods for Optimization of Discretized Continuous Functions. (in preparation).

Teaching

- Sep. 2023 – **Instructor**, *University of British Columbia, Vancouver, BC*
(Apr. 2026)
- Intro to Computing: Python (Fall 2025, Winter 2026)
 - Calculus I (Fall 2023, Winter 2025)
- Sep. 2022 – **Teaching Assistant**, *University of British Columbia, Vancouver, BC*
Dec. 2024
- Calculus I (Fall 2024)
 - Calculus II (Winter 2023)
 - Calculus III (Fall 2022)
- Sep. 2022 – **Mathematics Learning Centre Tutor**, *University of British Columbia,*
Apr. 2024 *Vancouver, BC*
- June 2020 – **Mathematics & Physics Tutor**, *Self-Employed*
Dec. 2024 *High School and Undergraduate Levels*
- Jan. 2018 – **Teaching Assistant**, *University of Waterloo, Waterloo, ON*
Dec. 2021
- Mathematics of Music (Fall 2020, 2021)
 - Special Topics in Mathematical Connections: Math & Music (Fall 2021)
 - Complex Analysis (Winter 2020)
 - Calculus 1 & 2 (Fall 2019, Winter 2018, 2020, 2021)

Awards

Society of Industrial and Applied Mathematics

July 2024 Student Travel Award

Natural Sciences and Engineering Research Council of Canada

May 2023 Canadian Graduate Scholarship – Doctoral (330 awards nationwide)

May 2021 Canadian Graduate Scholarship – Master's (840 awards nationwide)

May 2019 Undergraduate Student Research Award (3 149 awards nationwide)

University of British Columbia

May 2022 Faculty of Science PhD Tuition Award

Sep. 2022 Four Year Doctoral Fellowship

Sep. 2022 President's Academic Excellence Initiative PhD Award

University of Waterloo

- May 2021 President's Graduate Scholarship
- Aug. 2020 K.D. Fryer Gold Medal
- June 2020 Dean's Honours List With Distinction
- May 2019 President's Research Award
- Jan. 2019 Paul Berg Memorial Award
- Jan. 2019 Arthur Beaumont Memorial Scholarship
- Sep. 2016 President's Scholarship of Distinction
- Sep. 2016 Alumni @ Microsoft Entrance Scholarship in Mathematics

Software

Computer Skills

Julia, Python, MATLAB, Quarto, Typst, $\text{\LaTeX}$, Adobe Software, Haskell

Published Code & Packages

- 2024 MatrixTensorFactor.jl, *Julia*
github.com/MPF-Optimization-Laboratory/MatrixTensorFactor.jl
- 2023 SedimentSourceAnalysis.jl, *Julia*
github.com/njericha/SedimentSourceAnalysis.jl
- 2022 Sparse Random Mode Decomposition, *Python*
github.com/GiangTTran/SparseRandomModeDecomposition
- 2017 Drasil, *Haskell*
jacquescarette.github.io/Drasil

Conference Presentations & Posters

- July 2025 **Joint SIAM/CAIMS Annual Meetings, Montréal, QC**
BlockTensorDecomposition.jl: An Optimization Framework for Constrained Tensor Factorization, Pres.
- July. 2025 **CAIDA/TrustML Workshop — ICML Visitors @ UBC, Vancouver, BC**
Unsupervised Signal Demixing with Tensor Factorization, Poster.
- July. 2025 **Vision and Learning Workshop @ ICML, Vancouver, BC**
Unsupervised Signal Demixing with Tensor Factorization, Poster.
- Dec. 2024 **CMS Winter Meeting, Richmond, BC**
Density Separation with Tensor Factorization, Pres.
- Oct. 2024 **INFORMS Annual Meeting, Seattle, WA**
Density Separation with Tensor Factorization, Pres.
- Sep. 2024 **West Coast Optimization Meeting, Vancouver, BC**
Density Separation with Tensor Factorization, Pres.

- July 2024 **SIAM Annual Meeting**, *Spokane, WA*
Density Separation with Tensor Factorization, Pres.
- Oct. 2023 **Biennial Meeting of SIAM Pacific Northwest Section**, *Bellingham, WA*
Non-negative Tensor Decompositions for Unsupervised Learning, Pres.
- June 2021 **Canadian Applied and Industrial Mathematics Society**, *Waterloo, ON*
Sparse Random Wavelet Representation & Decomposition, Poster.
Runner up in annual poster competition

Other Research

- May 2019 – **Computational Fluids Research Assistant**, *University of Waterloo*,
Aug. 2019 Waterloo, ON, supervised by Michael Waite
- May 2017 – **Software Engineering Research Assistant**, *McMaster University*,
Aug. 2017 Hamilton, ON, supervised by Spencer Smith & Daniel Szymczak

Community & Leadership

- Sep. 2023 – **Math Graduate Committee**, University of British Columbia
(Apr. 2026) American Mathematical Society Graduate Student Chapter
 - Co-organizer and Seminar Host (2024–)
 - Social Convenor (2023–)
- Sep. 2022 – **UBC A Cappella Club**, Vancouver, BC
(Apr. 2026)
 - Live and Recording Sound Engineer (2024–)
 - Arranger (2023–)
- Sep. 2016 – **UW A Cappella Club**, Waterloo, ON
Apr. 2022
 - Arranger (2017–2022)
 - Musical Director (2017–2020)
 - Chair (2018–2020)
Club President Award and *Best Club Award*
(Waterloo Undergraduate Student Association, 2020)
- Oct. 2018 – **Canadian A Cappella Conference**, Waterloo, ON and Virtual
- Oct. 2021 Annual conference of workshops with over 100 singers from across the country
 - Lead Organizer (2019–2021)
Best Club Initiative
(Waterloo Undergraduate Student Association, 2019)
 - Travel Organizer (2018)